

Lecture 12-13 - Congestion Control ANS

1. What primary problem did the 1980s Internet face before TCP implemented congestion control?

- A. Low link utilization due to conservative sending rates
- B. Congestion collapse from hosts retransmitting aggressively after loss
- C. Frequent routing loops caused by distance-vector instability
- D. Packet corruption dominating over packet loss

ANS:

2. In the router-queue model from Lecture 12, what happens to average packet delay as load approaches 100%?

- A. Delay stays constant while loss increases linearly
- B. Delay decreases because queues drain faster
- C. Delay grows rapidly even before hitting 100% load
- D. Delay is unaffected by load in a FIFO queue

ANS:

3. Why is congestion control fundamentally described as a resource allocation problem?

- A. It chooses which routing paths packets follow
- B. It determines how bandwidth is divided among competing connections
- C. It decides which packets must be encrypted for security
- D. It configures router buffer sizes at startup

ANS:

4. Which of the following is NOT listed as a design goal for a congestion control algorithm?

- A. Avoid congestion by minimizing delay and loss
- B. Maximize link utilization (efficiency)
- C. Ensure fairness among connections
- D. Guarantee zero packet loss under all conditions

ANS:

5. Why is the "dynamic adjustment" approach especially suitable for the deployed Internet?

- A. It assumes routers coordinate globally using a central controller
- B. It requires no prior knowledge of available bandwidth or payment model
- C. It relies on per-flow reservations at every router
- D. It eliminates the need for flow control at receivers

ANS:

6. In the host-based dynamic adjustment approach, which three components must every sender implement?

- A. Route discovery, header compression, and packet scheduling
- B. Initial-rate discovery, congestion detection, and rate increase/decrease rules
- C. Fair queuing, ECN marking, and bufferbloat detection
- D. Error-correcting codes, flow labeling, and ACK pacing

ANS:

7. What signal does classic TCP use as its primary indication of congestion?

- A. Explicit ECN bits set by routers
- B. Growth of RTT beyond a threshold
- C. Packet loss detected via missing ACKs or receiving multiple duplicate ACKs
- D. Application-level throughput drops

ANS:

8. Which is a key downside of using delay-based congestion detection compared to loss-based detection?

- A. Delay-based schemes cannot react to congestion
- B. Packet delay can vary for reasons other than queue buildup
- C. Delay measurements require changes to packet headers
- D. Delay-based schemes cannot support high-speed links

ANS:

9. What is the main purpose of TCP slow start?

- A. Keep the window small to minimize queuing delay
- B. Quickly discover an approximate available bandwidth starting from a low sending rate
- C. Guarantee fairness between flows of different RTTs
- D. Prevent all packet loss during connection startup

ANS:

10. In slow start, what rate-adjustment behavior causes the window to grow exponentially?

- A. Adding a fixed number of packets to CWND each RTT
- B. Halving CWND on every ACK
- C. Doubling CWND every RTT by adding one packet per ACK in that RTT
- D. Keeping CWND constant and only adjusting the timeout

ANS:

11. What distinguishes AIMD from AIAD, MIAD, and MIMD in the fairness analysis?

- A. AIMD always keeps CWND fixed while the others change it
- B. AIMD is the only scheme that converges to a fair allocation
- C. AIMD guarantees zero loss while others do not
- D. AIMD is the only scheme that can be implemented at routers

ANS:

12. In the two-flow rate-adjustment graph, what does the "efficiency line" $X + Y = C$ represent?

- A. The set of allocations where one flow gets all bandwidth
- B. Allocations where total used bandwidth equals link capacity (full utilization of the link)
- C. Allocations with minimal delay and no loss
- D. Allocations that are guaranteed fair

ANS:

13. What is a major benefit of Explicit Congestion Notification (ECN) over pure loss-based congestion signaling?

- A. It guarantees zero queueing delay on all links
- B. It lets routers warn senders about congestion before queues overflow and packets are dropped
- C. It removes the need for ACKs entirely
- D. It makes TCP throughput completely independent of RTT

ANS:

14. Why can opening many parallel TCP connections be considered a cheating strategy?

- A. Because TCP shares bandwidth per host rather than per connection, so multiple connections are treated as one
- B. Because each connection receives its own share of bandwidth, so a user with many connections captures more total bandwidth than a user with one
- C. Because routers automatically deprioritize packets from users with multiple simultaneous flows
- D. Because duplicate ACK detection is disabled across parallel connections

ANS:

15. In AIMD, what happens to the difference between two flows' rates when both decrease on loss?

- A. The difference doubles
- B. The difference stays the same
- C. The difference halves
- D. The difference becomes zero immediately

ANS:

16. In the event-driven TCP model, which events cause CWND updates?

- A. New ACKs only
- B. New ACKs and duplicate ACKs only
- C. New ACKs, duplicate ACKs, and timeout events
- D. Routing updates and ARP replies

ANS:

17. During event-driven slow start, why does CWND effectively double every RTT?

- A. The sender doubles CWND whenever a timeout occurs
- B. Each ACK both frees one packet in the window and increments CWND by one, allowing two packets to be sent per ACK
- C. The receiver explicitly sends its desired window size each RTT
- D. The router increases the rate field in packet headers

ANS:

18. What role does SSTHRESH play in TCP congestion control?

- A. It stores the maximum allowed RTT

- B. It records the last observed loss rate
- C. It remembers a "safe" window size where slow start should stop and AIMD should begin
- D. It indicates the number of duplicate ACKs seen so far

ANS:

19. In the fast-recovery example, what is the main problem with naive CWND halving after an isolated loss?

- A. It causes persistent routing loops
- B. It forces the sender to stop sending until the retransmission is fully acknowledged
- C. It increases the number of corrupt packets
- D. It makes duplicate ACKs impossible

ANS:

20. What key idea underlies fast recovery in TCP Reno/New Reno?

- A. Use ECN bits to avoid all packet loss
- B. Grant temporary "credit" for each duplicate ACK to keep packets in flight
- C. Reset CWND to 1 packet after every loss
- D. Move congestion control into routers instead of hosts

ANS:

21. In the combined TCP-with-congestion-control logic, what happens on three duplicate ACKs?

- A. CWND is set to 1 packet and slow start restarts
- B. Ssthresh is set to $\text{CWND}/2$ and CWND is set to $\text{CWND}/2 + 3$, then the leftmost unacked packet is retransmitted
- C. RTT estimation is reset and no retransmission is done
- D. The receiver increases its advertised window

ANS:

22. What is a major benefit of Explicit Congestion Notification (ECN) over pure loss-based congestion signaling?

- A. It guarantees zero queueing delay on all links
- B. It lets routers warn senders about congestion before queues overflow and packets are dropped
- C. It removes the need for ACKs entirely
- D. It makes TCP throughput completely independent of RTT

ANS:

23. What simplifying assumption is made in the TCP throughput model about loss events?

- A. Losses are random and independent of window size
- B. Exactly one packet is lost whenever CWND exceeds a fixed $W_{\max}$
- C. Loss always happens in bursts of multiple packets
- D. Loss is caused only by link corruption, never congestion

ANS:

24. Under the derived TCP throughput model, how does average window size relate to $W_{\max}$?

- A. It equals $W_{\max}$
- B. It equals $W_{\max}/2$
- C. It equals $3/4 \cdot W_{\max}$
- D. It equals $2 \cdot W_{\max}$

ANS:

25. What key qualitative conclusion does the TCP throughput equation draw about flows with different RTTs?

- A. Throughput is independent of RTT, so all flows are RTT-fair
- B. Throughput is inversely proportional to RTT, so shorter RTT flows get more bandwidth
- C. Throughput is proportional to RTT squared
- D. RTT only affects slow start, not steady-state throughput

ANS:

26. Why do short TCP connections often suffer higher latency than long ones?

- A. They always traverse more congested paths
- B. They spend most of their lifetime in slow start and may see timeouts instead of fast retransmit due to duplicate ACKs
- C. Routers deprioritize short flows by default
- D. They cannot use acknowledgments for reliability

ANS:

27. What problem does bufferbloat create in TCP networks?

- A. Extremely low utilization because queues are too small
- B. Huge queues that add latency before loss signals arrive
- C. Inability to distinguish ACKs from data packets
- D. Excessive corruption due to noisy links

ANS:

28. Which property of fair queuing is highlighted as a benefit over simple FIFO queues?

- A. It guarantees zero congestion on all links
- B. It isolates flows so cheaters and RTT differences have less impact on others
- C. It removes the need for congestion control at end hosts
- D. It is simpler to implement than FIFO

ANS:

29. What is the core idea behind Explicit Congestion Notification (ECN)?

- A. Routers silently drop packets earlier than usual
- B. Routers set a bit in packet headers to warn of impending congestion before loss
- C. Senders mark packets to request higher priority
- D. Receivers estimate congestion by measuring jitter

ANS:

30. Why are congestion control and reliability described as "intertwined" in classic TCP?

- A. Both functions are implemented in routers only
- B. Congestion windows, ACKs, and timeouts are used simultaneously for rate control and detecting loss
- C. Reliability is implemented at the application layer but depends on router ECN
- D. Congestion control requires forward error correction instead of retransmission

ANS:

31. Why can opening many parallel TCP connections be considered a cheating strategy?

- A. Because TCP shares bandwidth per host rather than per connection, so multiple connections are treated as one
- B. Because each connection receives its own share of bandwidth, so a user with many connections captures more total bandwidth than a user with one
- C. Because routers automatically deprioritize packets from users with multiple simultaneous flows
- D. Because duplicate ACK detection is disabled across parallel connections

ANS:

32. In the max-min fairness example with total capacity $C = 10$ and demands $A = 8$, $B = 6$, $C = 2$, what is the correct max-min fair allocation?

- A. $A = 5$, $B = 3$, $C = 2$
- B. $A = 4$, $B = 4$, $C = 2$
- C. $A = 6$, $B = 2$, $C = 2$
- D. $A = 8$, $B = 1$, $C = 1$

ANS: